

# SurvivorLAN

When Alexandria falls, the network must survive.



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# ABOUT US



Yousef Hazaa



Nadein Reda



Youssef Aboolo



Youssef Ehab



Marwan Gabrallah

Mostafa Eltorky

# PROJECT OVERVIEW

Imagine Alexandria during a zombie apocalypse...

Communication is critical, but the network is under attack.

Our mission is to build a secure, segmented, and resilient network that keeps critical departments connected while keeping the Zombies isolated.

Our Mission:

- Connect all critical departments securely.
- Separate networks using VLANs.
- Control communication using Routing & ACLs.
- Protect critical systems from the Zombie Network.
- Monitor and demonstrate the network's security and availability.

“Survive the attack. Secure the network. Keep Alexandria connected.”

# PROJECT OBJECTIVES

What Are We Trying to Achieve?

Our goal is to design a secure and segmented network that keeps Alexandria's critical services operational during a simulated Zombie Apocalypse.

## Key Objectives

- Network Segmentation — Separate departments using VLANs.
- Secure Communication — Control traffic between departments.
- Zombie Isolation — Prevent unauthorized access to critical networks.
- Reliable Connectivity — Maintain communication between authorized departments.
- Network Services — Provide essential services such as DHCP, DNS, HTTP, and FTP.
- Monitoring & Response — Detect, block, and troubleshoot network issues.

Mission:

Keep the critical infrastructure alive while keeping the Zombies out.

# SurvivorLAN network architecture

Our network is divided into isolated segments, with each department assigned to a dedicated VLAN

## Core Technologies

- VLANs
- Trunking
- Inter-VLAN Routing
- OSPF
- ACLs

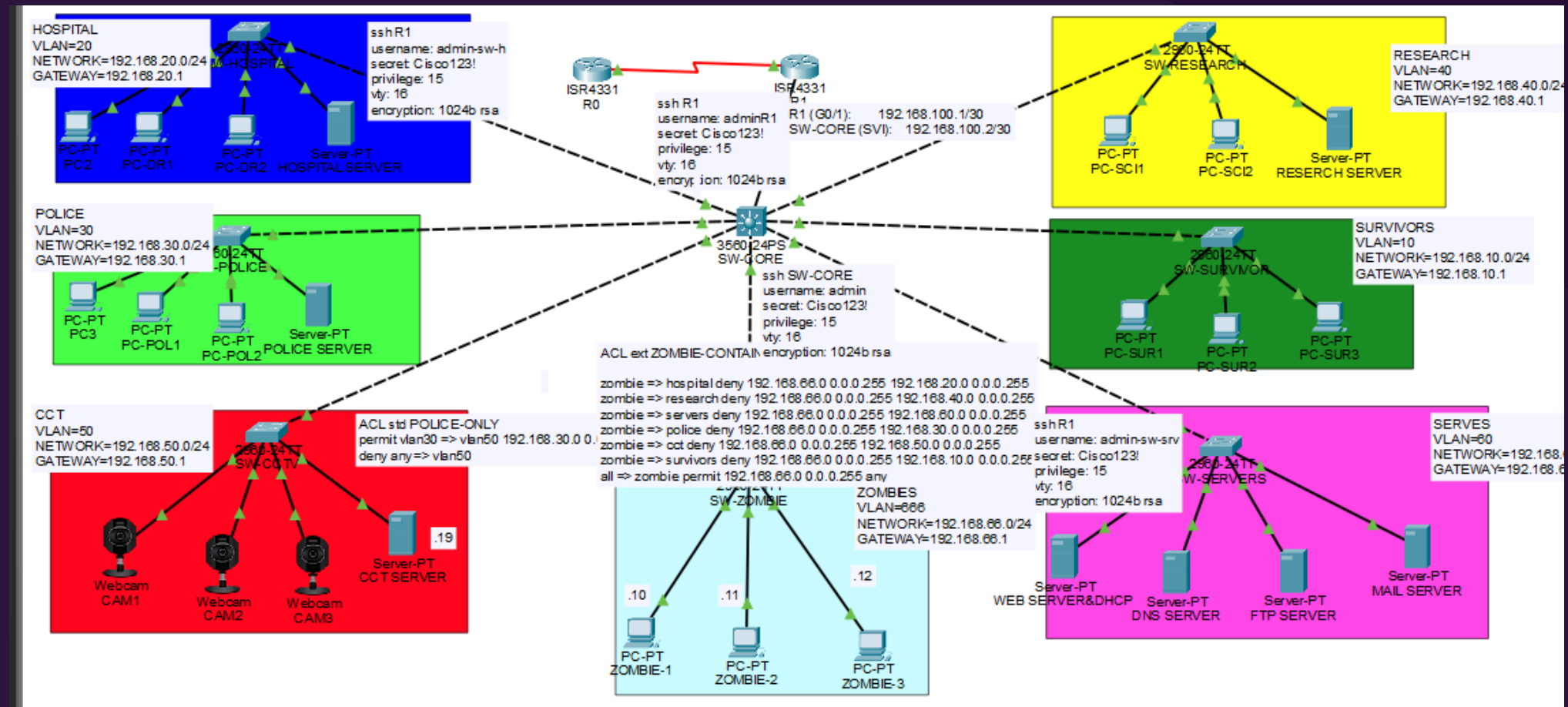
| VLAN | Department   | Network         |
|------|--------------|-----------------|
| 10   | Survivors    | 192.168.10.0/24 |
| 20   | Hospital     | 192.168.20.0/24 |
| 30   | Police       | 192.168.30.0/24 |
| 40   | Research Lab | 192.168.40.0/24 |
| 50   | CCTV         | 192.168.50.0/24 |
| 60   | Servers      | 192.168.60.0/24 |
| 665  | Zombies      | 192.168.66.0/24 |

# NETWORK TOPOLOGY

The network follows a centralized architecture, where the Core Network connects all critical departments and network services.

## Core Infrastructure

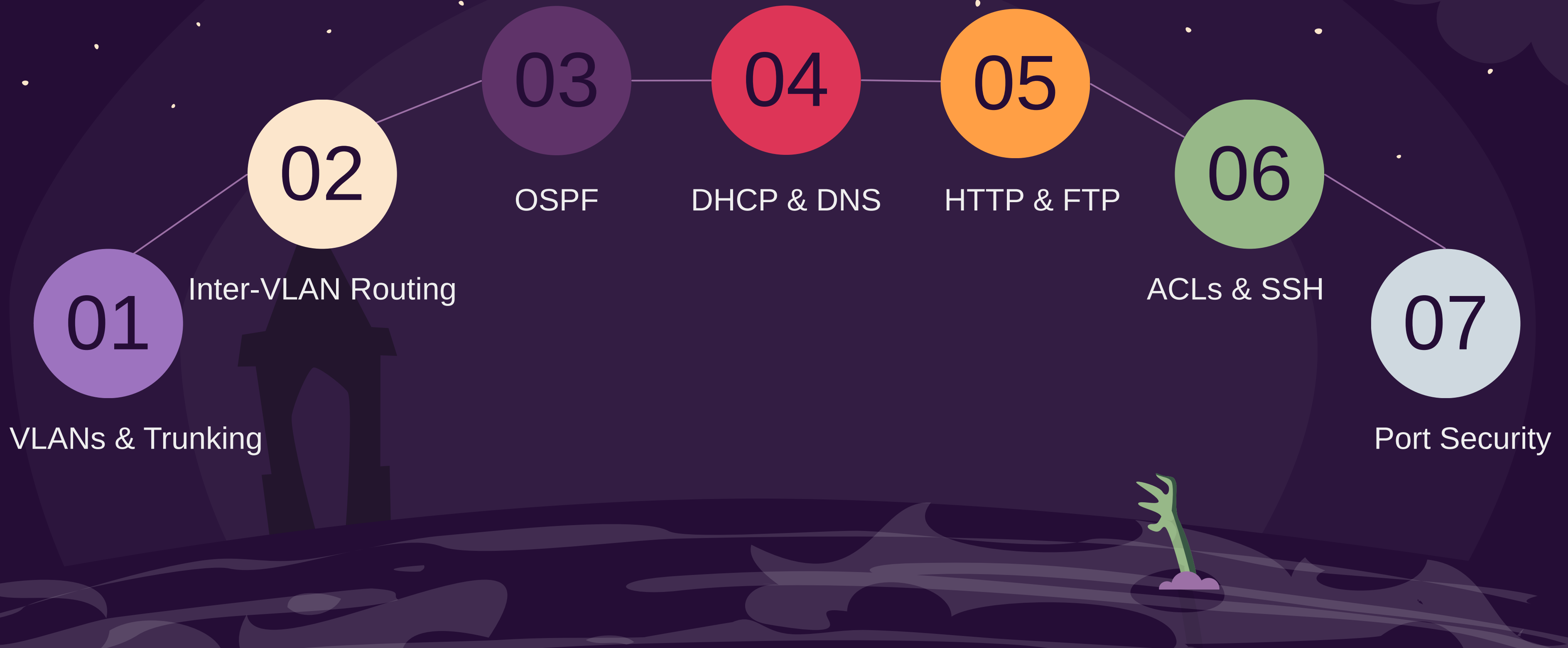
- Core Router — Connects the internal network to external networks.
- Core Switch — Central point for VLANs and inter-VLAN communication.
- Department Switches — Connect each department to its dedicated VLAN.
- Server Network — Hosts essential network services.
- Zombie Network — Separate untrusted segment assigned to VLAN 666.



Centralized network architecture connecting all departments through the Core Network, with the Zombie Zone isolated as an untrusted segment.

# NETWORK TECHNOLOGIES

## Technologies Used



# ROUTING & CONNECTIVITY

Keeping Alexandria Connected

Our network uses Inter-VLAN Routing and OSPF to provide reliable communication between different network segments.

## Inter-VLAN Routing

Enables communication between authorized VLANs.



## Default Route

Provides a path to external networks and the Internet.



## OSPF

Provides dynamic routing between network devices.



## Connectivity Testing

Ping and routing checks are used to verify network reachability.



# NETWORK SERVICES

- **DHCP**

Automatically assigns IP addresses to devices.

- **HTTP / Web Server**

Hosts the Alexandria 404 — Survival Dashboard.

- **DNS**

Resolves domain names to IP addresses.

- **FTP**

Provides file transfer between authorized devices.

- **Email Server**

Supports communication between departments.



# CYBERSECURITY



## ACLs

Control traffic between networks and block unauthorized access.



## Zombie Isolation

VLAN 666 is restricted from accessing critical departments.



## Port Security

Limits unauthorized devices on switch ports.



## SSH

Provides secure remote management of network devices.

Security Principle

Allow Authorized Traffic → Block Unauthorized Traffic

“The Zombies can attack. They  
just can't get in”



# ZOMBIE NETWORK ISOLATION

VLAN 666 — Untrusted Network:

Zombie Zone

ACL Protection

Critical Networks Blocked

Blocked Access

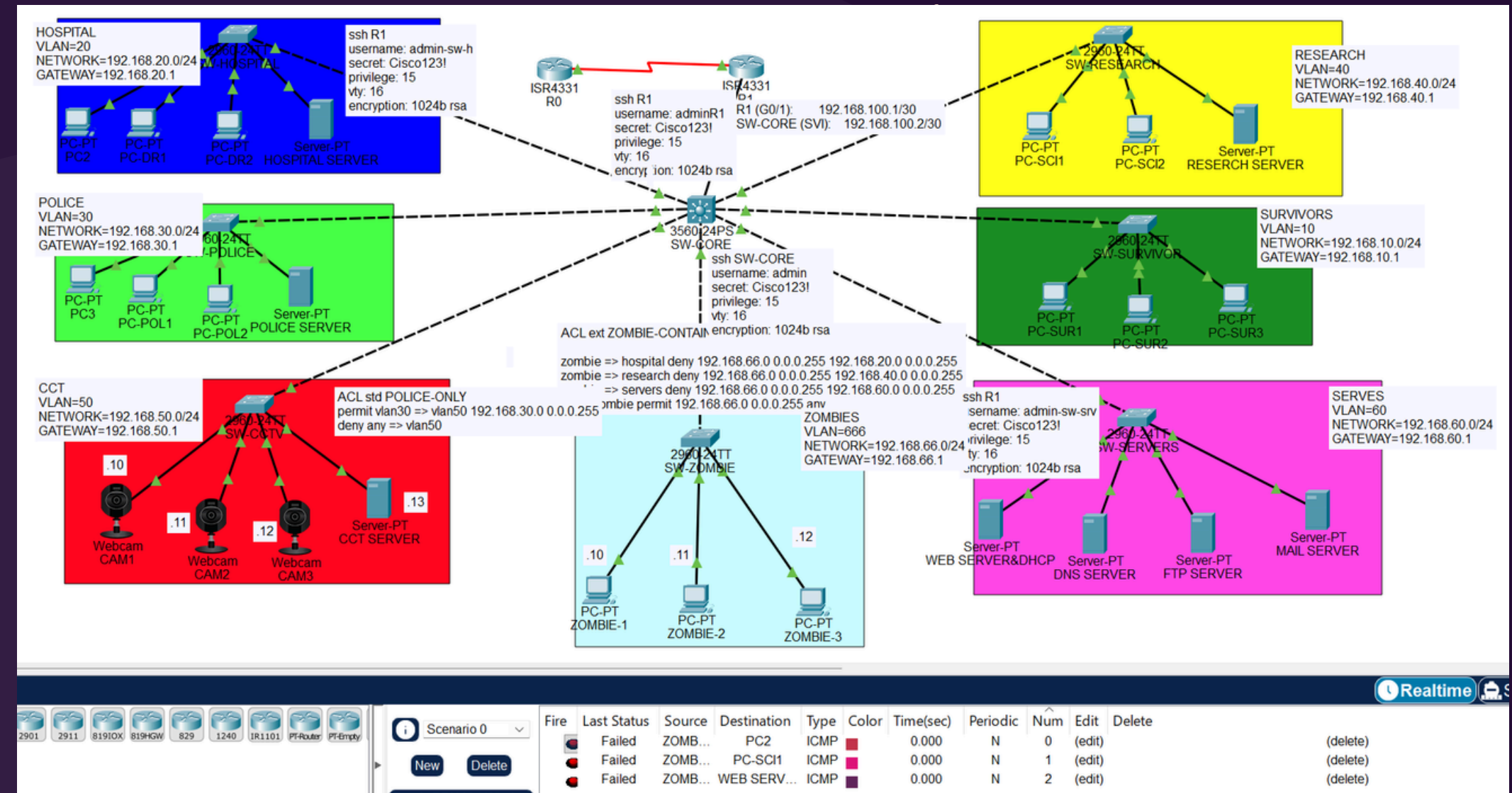
Hospital → BLOCKED

Research Lab → BLOCKED

Servers → BLOCKED

SUVIVORS → BLOCKED

POLICE → BLOCKED



```
SW-CORE#show access-lists POLICE-ONLY
Standard IP access list POLICE-ONLY
 permit 192.168.30.0 0.0.0.255 (1 match(es))
 deny any (6 match(es))

SW-CORE#show access-lists ZOMBIE-CONTAINMENT
Extended IP access list ZOMBIE-CONTAINMENT
 deny ip 192.168.66.0 0.0.0.255 192.168.20.0 0.0.0.255
 deny ip 192.168.66.0 0.0.0.255 192.168.40.0 0.0.0.255
 deny ip 192.168.66.0 0.0.0.255 192.168.60.0 0.0.0.255
 deny ip 192.168.66.0 0.0.0.255 192.168.30.0 0.0.0.255
 deny ip 192.168.66.0 0.0.0.255 192.168.10.0 0.0.0.255 (2 match(es))
 permit ip any 192.168.66.0 0.0.0.255
```

Zombie PC → Hospital Server

Request Timed Out

ACL successfully blocked unauthorized traffic.

# SURVIVORLAN WEB DASHBOARD

## Alexandria 404 — Survival Network

A centralized dashboard for monitoring network health, security, and threats.

- Network Status
- Threat Center
- Security Status
- VLAN Monitoring
- Incident Simulation
- Response Monitoring

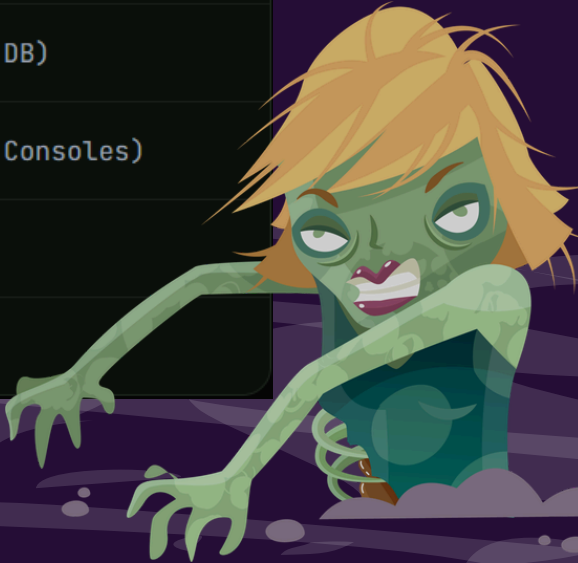
ACCESS THE LIVE DASHBOARD  
SURVIVORLAN — SURVIVAL NETWORK  
[ [Click to Open Dashboard](#) ]



# THREAT & SECURITY MONITORING

- Monitoring & Access Control
- The dashboard provides visibility into authorized and blocked network traffic.
- Authorized Flows
- Approved communication between critical VLANs.
- Blocked Traffic
- Unauthorized access attempts are denied.
- Zombie Isolation
- VLAN 666 is explicitly blocked from critical networks.
- Security Level
- Critical networks are continuously monitored.

| VLAN ID  | SECTOR                   | SUBNET        | SECURITY LEVEL        | NODES | STATE       | ACCESS RULE                                |
|----------|--------------------------|---------------|-----------------------|-------|-------------|--------------------------------------------|
| VLAN 10  | Survivor Zone            | 10.10.10.0/24 | Low                   | 142   | Operational | Permit: VLAN 60 (Servers)                  |
| VLAN 20  | Hospital & Triage        | 10.10.20.0/24 | Critical              | 88    | Operational | Permit: VLAN 60 (Medical Telemetry DB)     |
| VLAN 30  | Police & Defense Command | 10.10.30.0/24 | High                  | 64    | Operational | Permit: VLAN 50 (CCTV & Turrets)           |
| VLAN 40  | Bio-Research Lab         | 10.10.40.0/24 | Critical              | 35    | Operational | Permit: VLAN 60 (Pathogen Genome DB)       |
| VLAN 50  | Perimeter CCTV & Sensors | 10.10.50.0/23 | High                  | 210   | Operational | Permit: VLAN 30 (Defense Command Consoles) |
| VLAN 60  | Core Tactical Servers    | 10.10.60.0/24 | Critical              | 24    | Operational | Permit: VLAN 20 (Hospital)                 |
| VLAN 666 | Infected / Zombie Zone   | 10.66.66.0/20 | Hostile / Quarantined | 914   | Isolated    | TOTAL SINKHOLE ISOLATION                   |



# NETWORK RESILIENCE & INCIDENT RESPONSE

## Handling Network Failures

A structured troubleshooting process is used to detect, diagnose, and recover from network problems.

## Incident Response Process

**DETECT:** Identify the network problem.

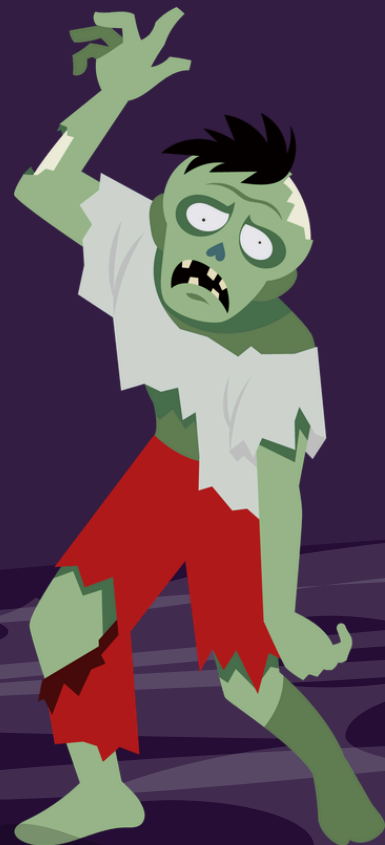
**DIAGNOSE:** Find the root cause using Cisco CLI commands.

**RECOVER:** Fix the affected connection or configuration.

**VERIFY:** Test connectivity and confirm network recovery.

## Key Troubleshooting Commands:

- show ip interface brief
- show vlan brief
- show ip route
- ping
- traceroute



# NETWORK RESILIENCE & INCIDENT RESPONSE

```
SW-CORE#show ip interface brief
Interface      IP-Address      OK? Method Status      Protocol
FastEthernet0/1 192.168.100.2   YES manual up          up
FastEthernet0/2 192.168.200.1   YES manual up          up
FastEthernet0/3 unassigned      YES unset up          up
FastEthernet0/4 unassigned      YES unset up          up
FastEthernet0/5 unassigned      YES unset up          up
FastEthernet0/6 unassigned      YES unset up          up
FastEthernet0/7 unassigned      YES unset up          up
FastEthernet0/8 unassigned      YES manual up          up
FastEthernet0/9 unassigned      YES unset down        down
FastEthernet0/10 unassigned      YES unset down        down
FastEthernet0/11 unassigned      YES unset down        down
FastEthernet0/12 unassigned      YES unset down        down
FastEthernet0/13 unassigned      YES unset down        down
FastEthernet0/14 unassigned      YES unset down        down
FastEthernet0/15 unassigned      YES unset down        down
FastEthernet0/16 unassigned      YES unset down        down
FastEthernet0/17 unassigned      YES unset down        down
FastEthernet0/18 unassigned      YES unset down        down
FastEthernet0/19 unassigned      YES unset down        down
FastEthernet0/20 unassigned      YES unset down        down
FastEthernet0/21 unassigned      YES unset down        down
FastEthernet0/22 unassigned      YES unset down        down
FastEthernet0/23 unassigned      YES unset down        down
FastEthernet0/24 unassigned      YES unset down        down
GigabitEthernet0/1 unassigned      YES unset down        down
GigabitEthernet0/2 unassigned      YES unset down        down
Vlan1          unassigned      YES unset administratively down down
Vlan10         192.168.10.1    YES manual up          up
Vlan20         192.168.20.1    YES manual up          up
Vlan30         192.168.30.1    YES manual up          up
Vlan40         192.168.40.1    YES manual up          up
Vlan50         192.168.50.1    YES manual up          up
Vlan60         192.168.60.1    YES manual up          up
Vlan666        192.168.66.1    YES manual up          up
```

```
R1#show ip route
```

```
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route
```

```
Gateway of last resort is 172.16.0.2 to network 0.0.0.0
```

```
172.16.0.0/16 is variably subnetted, 2 subnets, 2 masks
C    172.16.0.0/30 is directly connected, Serial0/2/0
L    172.16.0.1/32 is directly connected, Serial0/2/0
O    192.168.10.0/24 [110/2] via 192.168.100.2, 00:07:12, GigabitEthernet0/0/0
O    192.168.20.0/24 [110/2] via 192.168.100.2, 00:07:12, GigabitEthernet0/0/0
O    192.168.30.0/24 [110/2] via 192.168.100.2, 00:07:12, GigabitEthernet0/0/0
O    192.168.40.0/24 [110/2] via 192.168.100.2, 00:07:12, GigabitEthernet0/0/0
O    192.168.50.0/24 [110/2] via 192.168.100.2, 00:07:12, GigabitEthernet0/0/0
O    192.168.60.0/24 [110/2] via 192.168.100.2, 00:07:12, GigabitEthernet0/0/0
O    192.168.66.0/24 [110/3] via 192.168.100.2, 00:07:12, GigabitEthernet0/0/0
192.168.100.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.100.0/30 is directly connected, GigabitEthernet0/0/0
L    192.168.100.1/32 is directly connected, GigabitEthernet0/0/0
192.168.200.0/30 is subnetted, 1 subnets
O    192.168.200.0/30 [110/2] via 192.168.100.2, 00:07:12, GigabitEthernet0/0/0
S*   0.0.0.0/0 [1/0] via 172.16.0.2
```

```
R1#show ip ospf neighbor
```

| Neighbor ID   | Pri | State   | Dead Time | Address       | Interface            |
|---------------|-----|---------|-----------|---------------|----------------------|
| 192.168.200.1 | 1   | FULL/DR | 00:00:30  | 192.168.100.2 | GigabitEthernet0/0/0 |
| 172.16.0.2    | 0   | FULL/ - | 00:00:30  | 172.16.0.2    | Serial0/2/0          |



# TESTING & VALIDATION

## Verifying the Network

After integration, the complete network is tested to ensure that all components work correctly.

### Connectivity Tests:

- Ping Test

Verify communication between authorized networks.

- Routing Test

Check OSPF routes and network reachability.

- Service Test

Verify DHCP, DNS, HTTP, and FTP services.

- Security Test

Confirm that ACLs block unauthorized traffic.

- Zombie Isolation Test

Verify that VLAN 666 cannot access critical networks.

### Validation Result

Connectivity ☒

Services ☒

Security ☒

Isolation ☒

Tested. Verified. Ready for  
Deployment.



# PROJECT IMPACT

Our network design provides a secure and organized infrastructure for a city operating under extreme conditions.

## Key Benefits:

- Security

Critical departments are protected from unauthorized access.

- Reliability

Dynamic routing and network services maintain connectivity.

- Segmentation

Each department operates within its dedicated network.

- Threat Containment

The Zombie Zone is isolated from critical systems.

- Incident Readiness

The network supports monitoring, troubleshooting, and recovery.

- SURVIVORLAN

Secure the Network. Protect the Survivors



# CONCLUSION

SurvivorLAN — Survival Network

SurvivorLAN combines Networking, Security, Web Monitoring, and Troubleshooting into one integrated infrastructure designed for a zombie-apocalypse scenario.

Final Status:

-  Hospital —  OPERATIONAL
-  Police —  OPERATIONAL
-  Research Lab —  OPERATIONAL
-  Survivors —  OPERATIONAL
-  Servers —  OPERATIONAL
-  Zombie Zone —  ISOLATED

What We Achieved

Connected → Secured → Monitored → Tested → Recovered



# Thanks

Do you have any questions?

